

Periodic Table and Chemistry

The periodic table organizes elements by atomic number and recurring chemical properties.

Key facts:

Atomic number equals the number of protons in an atom's nucleus.

Elements in the same group often have similar valence electron patterns and chemical behavior.

Periods are rows in the periodic table and generally correspond to increasing electron shells.

Metals tend to conduct electricity and heat, while nonmetals vary widely in physical properties.

Ionic bonds often form through electron transfer between atoms, producing charged ions.

Covalent bonds involve atoms sharing pairs of electrons.

Chemical reactions rearrange atoms and bonds but do not create or destroy atoms in ordinary chemistry.

Answerable questions:

Q: What does atomic number mean? A: It is the number of protons in the nucleus of an atom.

Q: Why are elements grouped? A: Elements in the same group often share similar valence electron structures and chemical properties.